Monetary Policy Transmission and Building Land Scarcity: A Panel Local Projections Analysis of Regional Austrian Housing Markets

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Abstract

Acknowledgements

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1. Introduction

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2. Institutional and Economic Background

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4. Data and Variables

The empirical analysis rests on a balanced quarterly panel of Austrian districts covering the period from the first quarter of 2017 to the fourth quarter of 2025. The spatial unit is the political district, and the panel comprises 116 districts observed over 36 quarters, which yields 4,176 district-quarter observations. Vienna is retained at the level of its

individual municipal districts, so that the Viennese housing market enters the analysis at the same granularity as the remaining districts rather than as a single aggregate. The following subsections describe the outcome variable, the monetary policy shock, the supply-constraint measure, and the control variables in turn, before the final subsection sets out how these components are combined into the estimation panel.

4.1. Real Estate Price Data

Residential real estate prices are provided by Data Science Service GmbH, which compiles district-level price measures from transaction data. The data distinguish three property categories, average prices for single-family houses (EFH), average prices for condominium apartments (ETW), and building land prices, each expressed in euro per square metre. Building land is not treated as a housing outcome and is therefore excluded from the price series used in estimation. Over the full sample the price data are complete, with an observation for every district in every quarter and no missing values in any of the three categories.

The primary outcome is a district-level composite price constructed as a fixed-weight average of the house and apartment series. For each district, the composite price level is defined as $p_{i,t}^{\text{comp}} = w_i^{\text{house}} \, \text{EFH}_{i,t} + w_i^{\text{apt}} \, \text{ETW}_{i,t}$, where the weights are held constant over the sample and are derived from the district housing stock, as described in the next paragraph. Composing the series in levels and taking logarithms only afterwards avoids time-varying endogenous weights, while the fixed weights allow each district to be represented by the property mix that actually characterises its market. This choice motivates the property-specific robustness outcomes discussed in Section 7, where the house and apartment series are examined separately.

The composite weights are based on the building stock recorded in the 2011 Statistik Austria building and housing census, which reports the number of dwellings per district by the number of dwellings in the building. Dwellings in single-dwelling and two-dwelling buildings are assigned to the house category, the latter to capture semi-detached and duplex dwellings that behave like the owner-occupied house market, while dwellings in buildings with three or more dwellings are assigned to the apartment category. The weights are fixed at their 2011 values so that they predate the estimation window and remain exogenous to price developments during the sample, following the same logic that governs the supply-constraint measure below. Because the weights are time-invariant, cross-district composition is held fixed and only the underlying prices vary over time.

4.2. Monetary Policy Shocks

The monetary policy shock is constructed from the Euro Area Monetary Policy Event-Study Database of **altavilla2019<empty citation>**, which records high-frequency

changes in a range of asset prices within tight windows around ECB policy announcements. The chosen instrument is the surprise in the twenty-year overnight index swap rate (OIS 20Y). This maturity is selected because residential real estate prices depend on expected long-term financing conditions, so that surprises at the long end of the yield curve are the component of monetary policy most relevant for housing valuations. The two-year and thirty-year German government bond yield surprises are retained only as alternative instruments for robustness.

The event-level surprises are aggregated to the quarterly frequency of the panel by summing all surprises within a quarter into a net quarterly shock. Alongside the net measure, the sum of positive surprises, the sum of negative surprises, and the sum of absolute surprises are recorded for each quarter, which supports the signed decomposition used to examine asymmetric responses. The sign convention follows the interpretation of the underlying instrument, a negative OIS 20Y surprise denotes a downward revision of long-term rate expectations and therefore a monetary easing surprise. This convention matters for the interpretation of the estimated coefficients and is carried through consistently in Section 5. Over the sample the mean net quarterly surprise is negative, at roughly -0.59, consistent with the prevailing easing stance during much of the period, and the largest absolute net surprises occur in the third quarter of 2019 and around the onset of the tightening cycle in 2022 and 2023.

A preliminary diagnostic examines how far the quarterly net sum conceals offsetting surprises within a quarter. The cancellation ratio, defined as the absolute net shock divided by the sum of absolute surprises, has a mean of about 0.69, which indicates that easing and tightening surprises frequently occur within the same quarter and partly offset in the net measure. The ratio is a diagnostic rather than an additional regressor, and the extent of within-quarter cancellation motivates the alternative shock aggregations, such as the easing-only and tightening-only components, that are examined as robustness checks.

4.3. Supply-Constraint Measure

Regional differences in housing supply are captured through the district building land reserve recorded in the ÖROK monitoring of building land reserves (**oerok_blr**), which measures the share of vacant developable land in total designated building land. A lower reserve indicates less room for additional construction and therefore a more supply-inelastic district. To predate the estimation window and reduce concerns about endogeneity, the 2014 vintage of the reserve is used throughout, so that the classification cannot respond to price developments during the sample.

Districts are classified as supply-inelastic using the twenty-fifth percentile of the reserve distribution outside Vienna, a threshold of roughly 22 percent, which is then applied to all districts. This procedure classifies 47 districts as inelastic and 69 as elastic. Of the

inelastic districts, 24 lie outside Vienna and 23 are Viennese municipal districts, so that all Viennese districts and all Salzburg districts fall into the inelastic group. The measure is a proxy for supply conditions rather than a directly estimated supply elasticity, and it is strongly structured along geographic and compositional lines. In particular, the reserve measure correlates at about 0.91 with the fixed house weight, so that inelastic districts tend also to be apartment-heavy. District fixed effects absorb time-invariant level differences of this kind, but the entanglement of supply constraints with geography and property composition remains a concern for interpretation, which is why property-specific outcomes, a Vienna-excluded specification, and a continuous reserve measure are reported as robustness checks in Section 7.

4.4. Control Variables

The control set comprises district-level population, the unemployment rate, and net income per capita, each aligned to the quarterly frequency of the panel. Population is available quarterly from the central residence register and enters directly. Unemployment is available annually at the district level and is carried across the four quarters of each year. Net income per capita is constructed by dividing annual district net income totals by the corresponding population, so that within-year variation in income per capita partly reflects population movements. Since these series are annual in origin, the quarterly panel repeats the annual figures within each year, and this construction is reflected in the way the control lags behave.

Several series require construction at the tail of the sample, where source data are not yet available. Annual net income is observed through 2022 and is extrapolated for 2023 to 2025 using the nominal Tariflohnindex, so that all districts share a common wage-index growth path over the extrapolated years. District unemployment is observed through 2024 and carried forward to 2025, and population is carried forward from the first quarter of 2025 to the remaining quarters of that year. These constructed values are complete observations, but they do not represent newly observed district information. Their share is largest at the end of the sample and declines at longer horizons as later quarters drop out of the estimation window. To ensure that these choices do not drive the findings, the analysis reports a sensitivity sample that excludes the constructed tail.

Two variables considered at the design stage are not included. Tourism intensity is not available at the district level on a consistent basis over the sample period, since it is released by the federal states under differing conventions and frequencies, and it is therefore recorded as a data limitation rather than approximated with inconsistent series. Net migration, which features in the international literature, could be incorporated later should consistent annual district data become available, and population growth serves as the closest available proxy for local demand pressure in the meantime.

4.5. Final Panel Construction

The estimation panel is assembled by merging the price series, the composite weights, the supply-constraint classification, the quarterly shock, and the three control variables on the district and quarter keys. The price series and controls vary by district and quarter, the shock is common across districts within a quarter, and the supply classification and composite weights are time-invariant district characteristics. All component inputs join without duplicate or unmatched keys, and every district appears in every quarter, so the merged panel is balanced and complete at 4,176 observations. Districts are identified by a consistent numeric code across all sources, and the Viennese municipal districts are aligned to the three-digit district scheme used throughout the panel.

The dependent variable and the lagged controls introduce structural missing values by construction rather than through gaps in the source data. The cumulative outcome at horizon h is measured relative to the quarter preceding the shock, and four lagged quarterly changes enter as controls, so the usable sample declines from 3,596 observations at the shortest horizon to 2,204 at the longest. All 116 districts remain represented at every horizon, but each additional horizon removes the most recent quarter, so that comparisons of coefficients across horizons also compare progressively earlier samples. This feature is inherent to the local projection design and is disclosed alongside the estimates.

5. Empirical Strategy

5.1. Panel Local Projections

5.2. Baseline Model and Identification

5.3. Interpretation of the Coefficient

5.4. Outcome Transformation

5.5. Fixed Effects

5.6. Inference and Identification Assumptions

6. Descriptive Evidence

Before turning to estimation, this section documents the patterns in the assembled panel that motivate the empirical analysis. The evidence presented here is descriptive and is

not intended to support causal claims, since the differences reported below can reflect urbanity, dwelling composition, income, and other district characteristics rather than the transmission of monetary policy. The purpose is more modest, to establish that the data contain economically meaningful regional heterogeneity in price levels, price dynamics, and supply conditions, and to surface the features that shape the specification and interpretation of the local projections.

6.1. Prices and Regional Heterogeneity

Across the sample the composite price averages roughly 3,260 euro per square metre, with a standard deviation of about 1,590 and a range that spans from under 900 to over 13,000 euro per square metre. This wide dispersion is the first indication of substantial regional heterogeneity in Austrian housing markets. The house and apartment series differ markedly in level, with average house prices around 4,200 euro per square metre exceeding average apartment prices of about 2,870, and both series are right-skewed, so that a small number of high-price districts sit well above the median.

The dispersion is closely related to supply conditions. Grouping districts by the supply-constraint classification, the mean composite price in inelastic districts is roughly 4,410 euro per square metre against about 2,480 in elastic districts, so that the more supply-constrained districts operate in a distinctly higher price regime. As emphasised above, this level difference is not evidence of differential monetary transmission, since inelastic districts are also more urban and more apartment-heavy, and the reserve measure correlates strongly with the fixed house weight. It does, however, confirm that the supply classification separates districts that differ systematically in their housing markets, which is a precondition for the interaction analysis that follows.

6.2. Price Dynamics and Momentum

Quarterly composite price growth averages about 1.3 percent over the sample, with elastic districts growing marginally faster on average than inelastic districts. More important for the empirical strategy is the time-series behaviour of price growth. The pooled first-order autocorrelation of quarterly composite growth is about 0.23, and the autocorrelation remains positive at longer lags, which confirms the presence of the price momentum that is a well-documented feature of housing markets. This pattern is the direct empirical justification for including four lagged quarterly price changes among the controls, following **aastveit2018<empty citation>**. The same persistence, combined with the overlapping cumulative outcomes of the local projections, also foreshadows the serial-correlation concern that governs the treatment of standard errors discussed in Section 5.

6.3. Supply Constraints and Their Spatial Structure

The building land reserve varies considerably across districts, with a median of about 24 percent and values ranging from close to zero to above 45 percent. The supply classification is strongly clustered by federal state. All Viennese districts and all Salzburg districts fall into the inelastic group, while Burgenland contains no inelastic districts, and the median reserve in Vienna, at under 5 percent, is far below that of the predominantly rural states. This geographic concentration is economically intuitive, since the most supply-constrained districts are the dense urban and Alpine areas where developable land is genuinely scarce, but it also implies that the supply measure is entangled with region and with property composition. The fixed house weight and the reserve measure correlate at about 0.91, so that inelastic districts are also those in which apartments dominate the housing stock.

This entanglement carries two implications that are taken up in the empirical analysis. First, district fixed effects remove time-invariant differences in price levels and composition, but they do not by themselves rule out that districts differing in these correlated characteristics also respond differently to a common shock for reasons other than supply elasticity. Second, because the interaction with the shock is by construction non-zero only for inelastic districts, and because all Viennese districts are inelastic, the identifying variation is concentrated in a structurally distinctive set of markets. These considerations motivate the property-specific outcomes, the Vienna-excluded specification, and the continuous reserve measure that are reported as robustness checks, and they caution against reading the baseline interaction as a clean measure of supply elasticity alone.

6.4. The Monetary Policy Shock over the Sample

The quarterly net OIS 20Y surprise is predominantly negative over the sample, in line with the accommodative stance that characterised much of the period, before turning sharply positive around the onset of the tightening cycle in 2022. Both easing and tightening quarters are well represented, with an easing component present in 31 of the 36 quarters and a tightening component in 26, and both signs occurring within the same quarter in 21 quarters. This overlap is the source of the within-quarter cancellation documented in Section 4.2, and the presence of both directions of surprise is what makes the sample suitable for examining whether price responses are asymmetric across monetary policy phases. Because the shock is common to all districts within a quarter, its aggregate effect is absorbed by the time fixed effects, and the descriptive variation reported here bears on the identifying interaction only through its differential incidence across the supply groups.

7. Empirical Results

7.1. Baseline Results

7.2. Alternative Responses

7.3. Robustness Checks

8. Discussion

8.1. Interpretation of Findings

8.2. Relation to Literature

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A. Appendix